

-- Step 1 : Exploring Data

-- 1.1 :- View table structure (column names and data types)

DESCRIBE superstore_data;

Result Grid						
		Filter Rows:			Export:	Wrap Cell C
	Field	Type	Null	Key	Default	Extra
►	Row ID	int	YES		NULL	
	Order ID	text	YES		NULL	
	Order Date	text	YES		NULL	
	Ship Date	text	YES		NULL	
	Ship Mode	text	YES		NULL	
	Customer ID	text	YES		NULL	
	Customer Name	text	YES		NULL	
	Segment	text	YES		NULL	
	Country	text	YES		NULL	
	City	text	YES		NULL	
	State	text	YES		NULL	
	Postal Code	int	YES		NULL	
	Region	text	YES		NULL	
	Product ID	text	YES		NULL	
	Category	text	YES		NULL	
	Sub-Category	text	YES		NULL	
	Product Name	text	YES		NULL	
	Sales	double	YES		NULL	
	Quantity	int	YES		NULL	
	Discount	double	YES		NULL	
	Profit	double	YES		NULL	

-- 1.2 :- Display first 10 records from dataset

SELECT * FROM superstore_data LIMIT 10;

Result Grid										
Filter Rows: Export: Wrap Cell Content: Fetch rows:										
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	
1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	
2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	
3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	
4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	
5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	
6	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
7	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
8	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
9	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
10	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	

Filter Rows: Export: Wrap Cell Content: Fetch rows:												
City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit	
Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136	
Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.94	3	0	219.582	
Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.62	2	0	6.8714	
Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031	
Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2	2.5164	
Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Acces...	48.86	7	0	14.1694	
Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28	4	0	1.9656	
Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152	
Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by S...	18.504	3	0.2	5.7825	
Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	United States 3.6 Outlet Surge	114.9	5	0	34.47	

-- Insight

-- The dataset contains structured sales data with categorical and numerical fields.

-- The Order Date is stored as text, requiring conversion for time based analysis.

-- For better performance, date columns should ideally be stored in DATE format instead of TEXT.

-- Postal Code is stored as INT, which may lead to loss of leading zeros; TEXT format is preferable.

-- Step 2 : Filtering Data

-- 2.1 :- Filter data for West region

SELECT *

FROM superstore_data

WHERE Region = 'West'

limit 10;

Result Grid										
Filter Rows:										
Export: Wrap Cell Content: Fetch rows:										
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	
3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	
6	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
7	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
8	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
9	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
10	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
11	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
12	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
14	CA-2016-161389	12-05-2016	12-10-2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	
18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	

Filter Rows:												
Export: Wrap Cell Content: Fetch rows:												
City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit	
Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.62	2	0	6.8714	
Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Acces...	48.86	7	0	14.1694	
Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28	4	0	1.9656	
Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152	
Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by S...	18.504	3	0.2	5.7825	
Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.9	5	0	34.47	
Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092	
Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal black	911.424	4	0.2	68.3568	
Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine	407.976	3	0.2	132.5922	
West Jordan	Utah	84084	West	OFF-ST-10000107	Office Supplies	Storage	Fellowes Super Stor/Drawer	55.5	2	0	9.99	

-- 2.2 :- Filter data for Furniture category

SELECT *

FROM superstore_data

WHERE Category = 'Furniture';

Result Grid										
Filter Rows:										
Export: Export Wrap Cell Content: Wrap Fetch rows: Fetch										
	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City
▶	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson
	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale
	6	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
	11	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
	24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia
	25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem
	28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia
	30	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia
	37	CA-2016-117590	12-08-2016	12-10-2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson

Filter Rows:												
Export: Export Wrap Cell Content: Wrap Fetch rows: Fetch												
	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136
	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.94	3	0	219.582
	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031
	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Acces...	48.86	7	0	14.1694
	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092
	Philadelphia	Pennsylvania	19140	East	FUR-CH-10002774	Furniture	Chairs	Global Deluxe Stacking Chair, Gray	71.372	2	0.3	-1.0196
	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	1044.63	3	0	240.2649
	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases	Riverside Palais Royal Lawyers Bookcase, Royal...	3083.43	7	0.5	-1665.0522
	Philadelphia	Pennsylvania	19140	East	FUR-FU-10004848	Furniture	Furnishings	Howard Miller 13-3/4" Diameter Brushed Chrom...	124.2	3	0.2	15.525
	Richardson	Texas	75080	Central	FUR-FU-10003664	Furniture	Furnishings	Electrix Architect's Clamp-On Swing Arm Lamp, ...	190.92	5	0.6	-147.963

-- 2.3 :- Filter records with Sales greater than 1000

SELECT *

FROM superstore_data

WHERE Sales > 1000

limit 10;

Result Grid										
Filter Rows:		Export:		Wrap Cell Content:		Fetch rows:				
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	
11	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	
36	CA-2016-117590	12-08-2016	12-10-2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	
55	CA-2016-105816	12-11-2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United States	New York City	
68	CA-2014-106376	12-05-2014	12-10-2014	Standard Class	BS-11590	Brendan Sweed	Corporate	United States	Gilbert	
150	CA-2016-114489	12-05-2016	12-09-2016	Standard Class	JE-16165	Justin Ellison	Corporate	United States	Franklin	
166	CA-2014-139892	09-08-2014	09-12-2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	
168	CA-2014-139892	09-08-2014	09-12-2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	
216	CA-2015-146262	01-02-2015	01-09-2015	Standard Class	VW-21775	Victoria Wilson	Corporate	United States	Medina	

Filter Rows:		Export:		Wrap Cell Content:		Fetch rows:					
City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092
Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	1044.63	3	0	240.2649
Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases	Riverside Palais Royal Lawyers Bookcase, Royal...	3083.43	7	0.5	-1665.0522
Richardson	Texas	75080	Central	TEC-PH-10004977	Technology	Phones	GE 30524EE4	1097.544	7	0.2	123.4737
New York City	New York	10024	East	TEC-PH-10002447	Technology	Phones	AT&T CL83451 4-Handset Telephone	1029.95	5	0	298.6855
Gilbert	Arizona	85234	West	OFF-AR-10002671	Office Supplies	Art	Hunt BOSTON Model 1606 High-Volume Electric ...	1113.024	8	0.2	111.3024
Franklin	Wisconsin	53132	Central	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	1951.84	8	0	585.552
San Antonio	Texas	78207	Central	TEC-MA-10000822	Technology	Machines	Lexmark MX611dhe Monochrome Laser Printer	8159.952	8	0.4	-1359.992
San Antonio	Texas	78207	Central	FUR-CH-10004287	Furniture	Chairs	SAFCO Arco Folding Chair	1740.06	9	0.3	-24.858
Medina	Ohio	44256	East	TEC-MA-10000864	Technology	Machines	Cisco 9971 IP Video Phone Charcoal	1188	9	0.7	-950.4

-- 2.4 :- Filter data for a specific date range

SELECT *

FROM superstore_data

WHERE STR_TO_DATE('Order Date', '%m/%d/%Y')

BETWEEN '2016-01-01' AND '2016-12-31'

limit 10;

Result Grid										
Filter Rows:										
Export: Wrap Cell Content: Fetch rows:										
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	
26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	
43	CA-2016-101343	7/17/2016	7/22/2016	Standard Class	RA-19885	Ruben Ausman	Corporate	United States	Los Angeles	
48	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	
49	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	
56	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	
57	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	
58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	
59	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	
60	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	

Rows: Export: Wrap Cell Content: Fetch rows:												
City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit	
Los Angeles	California	90049	West	OFF-BI-10001634	Office Supplies	Binders	Wilson Jones Active Use Binders	11.648	2	0.2	4.2224	
Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories	Imation 8GB Mini TravelDrive USB 2.0 Flash Drive	90.57	3	0	11.7741	
Los Angeles	California	90049	West	OFF-ST-10003479	Office Supplies	Storage	Eldon Base for stackable storage shelf, platinum	77.88	2	0	3.894	
Dover	Delaware	19901	East	TEC-AC-10002167	Technology	Accessories	Imation 8gb Micro Traveldrive Usb 2.0 Flash Drive	45	3	0	4.95	
Dover	Delaware	19901	East	TEC-PH-10003988	Technology	Phones	LF Elite 3D Dazzle Designer Hard Case Cover, Lf...	21.8	2	0	6.104	
Troy	New York	12180	East	OFF-ST-10000604	Office Supplies	Storage	Home/Office Personal File Carts	208.56	6	0	52.14	
Troy	New York	12180	East	OFF-PA-10001569	Office Supplies	Paper	Xerox 232	32.4	5	0	15.552	
Troy	New York	12180	East	FUR-CH-10003968	Furniture	Chairs	Novimex Turbo Task Chair	319.41	5	0.1	7.098	
Troy	New York	12180	East	OFF-PA-10000587	Office Supplies	Paper	Array Parchment Paper, Assorted Colors	14.56	2	0	6.9888	
Troy	New York	12180	East	TEC-AC-10002167	Technology	Accessories	Imation 8gb Micro Traveldrive Usb 2.0 Flash Drive	30	2	0	3.3	

-- Note

-- STR_TO_DATE is used because Order Date is stored as text.

-- In real-world scenarios, storing dates in DATE format would improve performance.

-- Step 3 : Aggregation

-- 3.1 :- Total sales by each region

SELECT Region, SUM(Sales) AS Total_Sales

FROM superstore_data

GROUP BY Region;

Result Grid		
Filter Rows:		
Region	Total_Sales	
South	388983.58500000037	
West	713471.3445000004	
Central	497800.8728000007	
East	672194.0539999981	

-- Insight

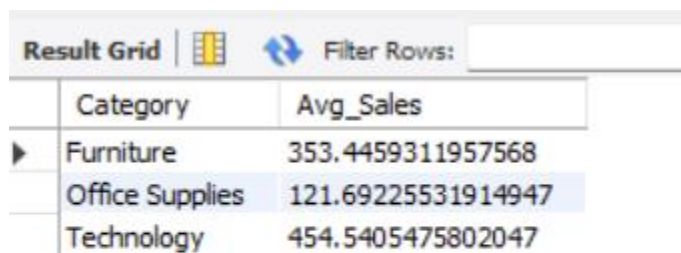
-- West and East regions contribute the highest sales, while the South region shows comparatively lower performance.

-- 3.2 :- Average sales by category

```
SELECT Category, AVG(Sales) AS Avg_Sales
```

```
FROM superstore_data
```

```
GROUP BY Category;
```



The screenshot shows a 'Result Grid' interface with a 'Filter Rows' input field. Below the header, there are three rows of data. The first row is 'Furniture' with an average sales value of 353.4459311957568. The second row is 'Office Supplies' with an average sales value of 121.69225531914947. The third row is 'Technology' with an average sales value of 454.5405475802047. The 'Office Supplies' row is highlighted in blue.

Category	Avg_Sales
Furniture	353.4459311957568
Office Supplies	121.69225531914947
Technology	454.5405475802047

-- Insight

-- Technology has the highest average sales, indicating fewer but high-value transactions.

-- Office Supplies has lower average sales, reflecting frequent low-cost purchases.

-- This suggests a difference in pricing strategy and customer buying behavior across categories.

-- 3.3 :- Total quantity sold by category

```
SELECT Category, SUM(Quantity) AS Total_Quantity
```

```
FROM superstore_data
```

```
GROUP BY Category;
```

Result Grid			Filter Rows:
	Category	Total_Quantity	
▶	Furniture	7855	
	Office Supplies	21990	
	Technology	6904	

-- Insight

-- Office Supplies has the highest quantity sold, indicating frequent purchases of these items.

-- Technology products have lower quantity but higher sales value per unit.

-- 3.4 :- Average quantity per order

```
SELECT AVG(order_qty)
```

```
FROM (
```

```
    SELECT `Order ID`, SUM(Quantity) AS order_qty
```

```
    FROM superstore_data
```

```
    GROUP BY `Order ID`
```

```
) t;
```

Result Grid		Filter Rows:
	AVG(order_qty)	
▶	7.4526	

-- Insight

-- The average quantity per order is low, indicating that most purchases are small-sized rather than bulk orders.

-- This suggests a retail-oriented customer base rather than wholesale buying behavior.

-- Step 4 : Sorting & Limiting

-- 4.1 :- Top 5 products by sales


```

SELECT `Product Name`, SUM(Sales) AS Total_Sales
FROM superstore_data
GROUP BY `Product Name`
ORDER BY Total_Sales DESC
LIMIT 5;

```

Result Grid  Filter Rows: Export:  Wrap Cell Con		
	Product Name	Total_Sales
▶	Canon imageCLASS 2200 Advanced Copier	61599.824
	Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.384
	Cisco TelePresence System EX90 Videoconferen...	22638.48
	HON 5400 Series Task Chairs for Big and Tall	21870.576
	GBC DocuBind TL300 Electric Binding System	19823.479000000003

-- Insight

-- A small number of products contribute significantly to total sales, showing a sales concentration pattern (Pareto effect).

-- High-value items like copiers dominate revenue contribution.

-- These products should be prioritized in inventory management and promotional strategies.

-- Step 5 : Business Use Cases (Core Analysis)

-- 5.1 :- Monthly Sales Trend (Time Analysis)

```

SELECT
    DATE_FORMAT(STR_TO_DATE(`Order Date`, '%m/%d/%Y'), '%Y-%m') AS Month,
    ROUND(SUM(Sales), 2) AS Total_Sales
FROM superstore_data
GROUP BY DATE_FORMAT(STR_TO_DATE(`Order Date`, '%m/%d/%Y'), '%Y-%m')
ORDER BY Month
limit 10;

```

Result Grid			Filter Rows:
	Month	Total_Sales	
▶	NULL	877699.44	
	2014-01	9254.18	
	2014-02	876.59	
	2014-03	47477.43	
	2014-04	12164.32	
	2014-05	14345.8	
	2014-06	17359.18	
	2014-07	24880.37	
	2014-08	11031.1	
	2014-09	48729.56	

-- Insight

-- Sales show seasonal peaks during the September–December period, especially in November, indicating strong festive demand.

-- A detailed year-over-year comparison would be required to confirm long-term growth trends.

-- 5.2 :- Top Customers (Revenue Contribution)

SELECT

 `Customer Name`,

 SUM(Sales) AS Total_Sales

FROM superstore_data

GROUP BY `Customer Name`

ORDER BY Total_Sales DESC

LIMIT 5;

Result Grid   Filter Rows:		
	Customer Name	Total_Sales
▶	Sean Miller	25043.05
	Tamara Chand	19017.847999999998
	Raymond Buch	15117.339
	Tom Ashbrook	14595.62
	Adrian Barton	14355.610999999997

-- Insight

-- A small group of customers contributes a significant share of total revenue.

-- This highlights the importance of customer retention strategies such as loyalty programs and personalized offers.

-- 5.3 :- Top Categories & Sub-Categories

SELECT

Category,

`Sub-Category`,

SUM(Sales) AS Total_Sales

FROM superstore_data

GROUP BY Category, `Sub-Category`

ORDER BY Total_Sales DESC;

Result Grid			
		Filter Rows:	Export: Wra
	Category	Sub-Category	Total_Sales
►	Technology	Phones	329753.0880000001
	Furniture	Chairs	328449.10300000076
	Office Supplies	Storage	216803.21200000012
	Furniture	Tables	206965.5320000001
	Office Supplies	Binders	199905.71700000006
	Technology	Machines	189238.63099999996
	Technology	Accessories	167380.3180000001
	Technology	Copiers	149528.02999999994
	Furniture	Bookcases	114879.99629999997
	Office Supplies	Appliances	107532.161
	Furniture	Furnishings	82752.23000000004
	Office Supplies	Paper	75356.11799999999
	Office Supplies	Supplies	45952.47000000001
	Office Supplies	Art	27118.791999999954
	Office Supplies	Envelopes	15339.489999999993
	Office Supplies	Labels	12486.312
	Office Supplies	Fasteners	3008.6559999999995

-- Insight

-- Sub-categories like Phones and Chairs generate the highest sales, indicating strong demand.

-- These segments are key revenue drivers and should be prioritized in inventory and marketing strategies.

-- 5.4 :- Duplicate Records Check (Data Quality)

```
SELECT `Order ID`, `Product Name`, COUNT(*) AS cnt
```

```
FROM superstore_data
```

```
GROUP BY `Order ID`, `Product Name`
```

```
HAVING COUNT(*) > 1;
```

Result Grid			
		Filter Rows:	
		Export:	
		Wrap Cell Content:	
	Order ID	Product Name	cnt
▶	CA-2016-129714	Xerox 1881	2
	US-2016-123750	Imation Secure + Hardware Encrypted USB 2.0 ...	2
	CA-2016-137043	Electrix Architect's Clamp-On Swing Arm Lamp, ...	2
	CA-2017-152912	Adjustable Depth Letter/Legal Cart	2
	US-2014-150119	Global Leather Highback Executive Chair with P...	2
	CA-2015-103135	GBC Prepunched Paper, 19-Hole, for Binding Sy...	2
	CA-2017-118017	Memorex Micro Travel Drive 16 GB	2
	CA-2016-140571	Xerox 1964	2

-- Insight

-- Duplicate records were identified based on Order ID and Product Name. Since this is a transactional dataset, multiple entries may represent valid repeated items within the same order rather than data errors.

-- 5.5 :- Shipping Delay Analysis

SELECT

 `Ship Mode`,

 AVG(DATEDIFF(

 STR_TO_DATE(`Ship Date`, '%m/%d/%Y'),

 STR_TO_DATE(`Order Date`, '%m/%d/%Y')

)) AS Avg_Shipping_Days

FROM superstore_data

GROUP BY `Ship Mode`

ORDER BY Avg_Shipping_Days;

Result Grid   Filter Rows:		
	Ship Mode	Avg_Shipping_Days
▶	Same Day	0.0208
	First Class	2.1719
	Second Class	3.1345
	Standard Class	4.9490

-- Insight

-- Shipping time varies across different ship modes. Faster shipping methods show lower delivery time, while economy modes take longer, helping evaluate logistics efficiency and service-level performance.

-- 5.6 :- Sales by Customer Segment

```
SELECT Segment, SUM(Sales) AS Total_Sales
```

```
FROM superstore_data
```

```
GROUP BY Segment
```

```
ORDER BY Total_Sales DESC;
```

Result Grid   Filter Rows:		
	Segment	Total_Sales
▶	Consumer	1150166.1819999903
	Corporate	696604.5138000002
	Home Office	425679.1605000003

-- Insight

-- Different customer segments contribute differently to revenue.

-- This helps identify the most valuable customer group for targeted strategies.

-- Profitability Analysis

SELECT

Category,

SUM(Sales) AS Total_Sales,

SUM(Profit) AS Total_Profit,

ROUND(SUM(Profit)/SUM(Sales)*100, 2) AS Profit_Margin

FROM superstore_data

GROUP BY Category

ORDER BY Profit_Margin DESC;

Result Grid					Filter Rows:	Export:	Wrap Cell Content:
	Category	Total_Sales	Total_Profit	Profit_Margin			
▶	Technology	835900.06699999964	145387.09659999999	17.39			
	Office Supplies	703502.92800000031	120489.88640000001	17.13			
	Furniture	733046.86129999996	16980.771199999998	2.32			

-- Step 6 : Data Validation (VERY IMPORTANT)

-- 6.1 :- Row Count Check

SELECT COUNT(*) FROM superstore_data;

Result Grid	
	COUNT(*)
▶	9694

-- Insight

-- The dataset contains 9694 records, which is adequate for exploratory analysis.

-- 6.2 :- Null Value Check

SELECT *

FROM superstore_data

WHERE

 `Row ID` IS NULL OR

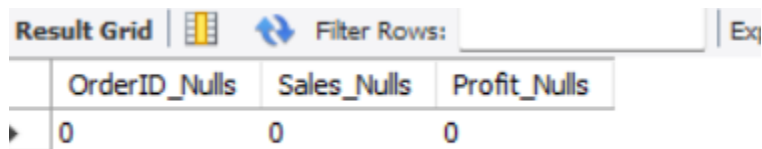
```

        `Order ID` IS NULL OR
`Order Date` IS NULL OR
`Ship Date` IS NULL OR
`Customer ID` IS NULL OR
`Customer Name` IS NULL OR
`Region` IS NULL OR
        `Category` IS NULL OR
        Sales IS NULL OR
        Quantity IS NULL OR
        Profit IS NULL;

SELECT

SUM(CASE WHEN `Order ID` IS NULL THEN 1 ELSE 0 END) AS OrderID_Nulls,
SUM(CASE WHEN Sales IS NULL THEN 1 ELSE 0 END) AS Sales_Nulls,
SUM(CASE WHEN Profit IS NULL THEN 1 ELSE 0 END) AS Profit_Nulls
FROM superstore_data;

```



	OrderID_Nulls	Sales_Nulls	Profit_Nulls
▶	0	0	0

```

-- Insight
-- No null values are returned for key columns in this check, indicating good data completeness.
-- However, a full-column null audit would ensure complete validation.

```

```

-- 6.3 :- Sales Consistency Check

```

```

SELECT MIN(Sales), MAX(Sales), AVG(Sales), STDDEV(Sales)
FROM superstore_data;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
MIN(Sales)	MAX(Sales)	AVG(Sales)	STDDEV(Sales)
0.444	22638.48	234.41818199917006	631.7564238264724

-- Insight

-- Sales values vary widely between minimum and maximum values, indicating a diverse range of transaction sizes.

-- However, further statistical analysis (such as standard deviation or box plots) is required to accurately identify outliers.

-- Additional Insight

-- The wide range of sales values reflects a mix of low-cost and high-value products.

-- 6.4 :- Negative Value Check

```
SELECT * FROM superstore_data
```

```
WHERE Sales < 0 OR Profit < 0;
```

```
SELECT
```

```
    COUNT(*) AS Loss_Making_Orders
```

```
FROM superstore_data
```

```
WHERE Profit < 0;
```

Result Grid	Filter Ro
Loss_Making_Orders	
1808	

-- Insight

-- A significant number of transactions show negative profit, indicating discounting, high costs, or inefficiencies that require further investigation.

-- Top 5 Cities by Sales

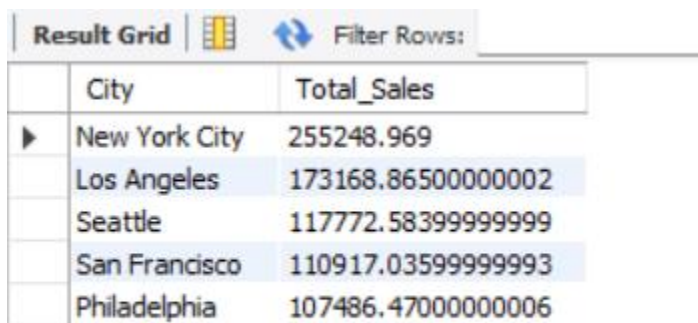
```
SELECT City, SUM(Sales) AS Total_Sales
```

```
FROM superstore_data
```

```
GROUP BY City
```

```
ORDER BY Total_Sales DESC
```

```
LIMIT 5;
```



The screenshot shows a database interface with a 'Result Grid' tab. It displays a table with two columns: 'City' and 'Total_Sales'. The table lists the top 5 cities by sales: New York City, Los Angeles, Seattle, San Francisco, and Philadelphia. The values for 'Total_Sales' are displayed with varying decimal precision.

	City	Total_Sales
▶	New York City	255248.969
	Los Angeles	173168.86500000002
	Seattle	117772.58399999999
	San Francisco	110917.03599999993
	Philadelphia	107486.47000000006

-- Insight

-- Sales are highly concentrated in major cities like New York and Los Angeles.

-- This indicates that urban markets are key revenue drivers, and expanding in similar high-density areas could increase sales.

-- Final Conclusion

-- The analysis shows that sales are driven by high-value products, key customers, and major urban regions, indicating a concentrated revenue distribution.

-- Technology products generate higher revenue per transaction, while Office Supplies drive volume.

-- Sales performance varies across regions, with West and East leading while South showing relatively lower contribution.

-- While the dataset is complete and suitable for analysis, deeper statistical methods can further enhance insights.